Laws of exponents

1 Simplify the following expressions:

a $a^s \times a^t$ **b** $\dfrac{a^s}{a^t}$ **c** $a^s \times a^s$ **d** $a^s + a^s$

e $a^s + a^t$ **f** $2^2 \times 2^2$ **g** $4y^3 \times 6y$ **h** $3x^5 \times 8x^2$

2 Simplify the following expressions:

a $\dfrac{x^6}{4x^4}$ **b** $p^{18} \div p^8 \div p^5$

c $\dfrac{4n^3 \times 4n^4}{16}$ **d** $\dfrac{u^{2x+1}}{u^x}$

e $p^7 \div p^3 \times p^5$ **f** $p^{14} \div p^6 \div p^5$

g $\dfrac{y^7 \times y^5}{y^3 \times y^2}$ **h** $12s^8 \div 3t^6 \times 6t^4$

i $\dfrac{10p^6 \times 3p^{10}}{15p^2}$ **j** $c^{2z}c^{z+1}$

k $20m^6 \div 5m^{13} \times 9m^2$ **l** $\dfrac{12y^4 \times 35y^8}{4y^5 \times 5y^4}$

3 Complete the statement below:

$$15j^{14} \div (\square) = 5j^7$$

4 Rewrite the following expressions without parantheses:

a $\left(\dfrac{a}{b}\right)^3$ **b** $(xy)^7$ **c** $(2x)^4$ **d** $\left(\dfrac{xy}{5z}\right)^2$

5 Write $(16^p)^4$ in the form a^b, where a is a prime number.

6 For each of the following expressions, state the operation that should be done to the exponents to simplify the expressions:

a $\left(z^{28}\right)^{19}$ **b** $z^{13} \times z^{16}$ **c** $z^{14} \div z^{13}$ **d** $\dfrac{z^{26}}{z^{13}}$

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7 Simplify the following expressions:

a $\dfrac{\left(x^3\right)^2}{x^3}$

b $\left(u^{x+1}\right)^3$

c $p^2 \times \left(p^3\right)^2$

d $p^8 \div \left(p^2\right)^2$

e $\dfrac{\left(x^2\right)^6}{\left(x^2\right)^2}$

f $\left(2y^4\right)^2 \times \left(2y^2\right)^3$

g $\dfrac{\left(4y^3\right)^5}{\left(y^2\right)^2}$

h $\left(3y^2\right)^3 \times \left(-2y^2\right)^2$

i $\left(u^9 \times u^5 \div u^{19}\right)^2$

j $\dfrac{28m^4 \times \left(3m^3\right)^3}{7m^2 \times \left(m^2\right)^2}$

k $\dfrac{\left(-35m^4\right) \times \left(3m^4\right)^3}{7m^3 \times \left(m^2\right)^5}$

l $\left(\dfrac{t^4}{t^2}\right)^5 \div \dfrac{t^5}{t^8}$

m $\left(t^6\right)^5 \div \left(\dfrac{t^9}{t^7}\right)^3$

n $\dfrac{\left(u^{x+3}\right)^3}{u^{x+1}}$

Zero and negative exponents

8 Simplify:

a a^0

b $8x^0$

c $(2 \times 13)^0$

d $\left(a^0\right)^{79}$

e $\left(13x^7\right)^0 + 13^0 - 13h^0$

f $\left(7m^0 + 4\right)^2$

9 Express the fraction $\dfrac{m}{n^4}$ as a product using negative exponents.

10 Express the following in simplest form without negative exponents:

a $4a^{-2}$

b $(3a)^{-1}$

c $6x^{-3}y^4$

d $m^2 \times m^{-7}$

e $m^{-5}n^{-4}p^4$

f $2y^9 \times 3y^{-5}$

g $\dfrac{9x^3}{3x^{-4}}$

h $\dfrac{12x^3}{4x^7}$

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11 Express the following in simplest form without negative exponents:

a $\left(\frac{a}{b}\right)^{-5}$ **b** $\left(4m^{-10}\right)^{4}$ **c** $\left(4m^{-8}\right)^{-3}$ **d** $\left(5y^{3}\right)^{-3}$

e $\left(\frac{x^{2}}{y^{4}}\right)^{-1}$ **f** $\left(\frac{a^{3}}{b^{3}}\right)^{-5}$ **g** $\left(\frac{2h}{3}\right)^{-4}$ **h** $\frac{\left(m^{-3}\right)^{-1} \times \left(m^{4}\right)^{-3}}{m^{3} \times m^{4}}$

12 Express the following in simplest form without negative exponents:

a $2p^{4}q^{-2} \times 5p^{-4}q^{-5}$ **b** $\frac{5p^{5}q^{-4}}{40p^{5}q^{6}}$

c $10x^{3}y^{2}z \div 2x^{5}y^{9}$ **d** $2^{15} \div 2^{-5}$

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